

QUOTIENT FAITHFULNESS IN REPRESENTATION LEARNING • PAPER V

Six Traditions, Three Failure Modes: The Admissibility Declaration as a Missing Structural Component in Representation Learning Theory

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Abstract

This paper is a perspective article on the place of the quotient-faithfulness framework within representation learning theory. It does not introduce new theorems. Its aim is conceptual positioning: situating the framework developed in the companion papers within the representation learning landscape, clarifying what each of six prior traditions addresses and what it leaves implicit, and generating falsifiable predictions that distinguish the framework from specific competing accounts.

The central interpretive claim is that several established traditions—Geometric Deep Learning, the Information Bottleneck, self-supervised anti-collapse methods, causal representation learning, disentangled representation learning, and the flat-minima tradition—each isolate genuine parts of the same broader problem but typically leave implicit the question of which equivalence relation a representation is meant to respect. From that perspective, an explicit admissibility declaration can be read as a missing structural component in each tradition: not a replacement for what those traditions have established, but a formal articulation of what they each leave underspecified. The framework map developed in the paper shows in each case where that gap lies and what the declaration adds.

The paper then revisits fourteen bodies of empirical evidence through that lens. These reframings are presented as interpretive comparisons with varying evidential strength, not as derived consequences of the formal results. Where a connection is conjectural, partial, or suggestive rather than tight, the paper says so explicitly through a five-tier evidence labelling scheme. Along the way the paper offers a mechanistic account of the Platonic Representation Hypothesis, distinguishes dimensional from semantic collapse in self-supervised learning, and derives three predictions that distinguish the framework from specific established competing accounts. A concluding section states the conditions under which each major practical claim would require revision.

The intended contribution is conceptual rather than technical. The paper argues that admissibility declarations provide a useful missing structural component in some existing discussions of compression, collapse, and invariance, while acknowledging that those traditions address aspects of the learning problem the current framework does not.

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series Quotient Faithfulness in Representation Learning develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural manifestations of criterion silence prevent standard training from achieving this: the embedding floor, graded compression, and quotient-respect failure.

Paper I [Farkas, 2026a]: Establishes that predictive loss does not certify quotient faithfulness; develops the embedding-space program and the three structural manifestations.

Paper II [Farkas, 2026b]: Identifies recovery routes and admissibility sources. Establishes the graded-compression structure and its Kolmogorov complexity foundations.

Paper III [Farkas, 2026c]: Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding in the single-quotient setting.

Paper IV [Farkas, 2026d]: Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers.

Paper V [Farkas, 2026e] (this paper): Maps the framework against representation-learning traditions and examines fourteen bodies of experimental evidence through the embedding-space lens.

Paper VI [Farkas, 2026f]: Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.

Paper VII [Farkas, 2026g]: Empirical validation on benchmark dynamical systems (phase cylinder, Hopf fibration, double pendulum) and an industrial case study (Color Fantasy ship-hotel energy system). Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].

Contents

Notation	4
1 Introduction and Reading Guide	5
1.1 The thesis	5
1.2 What this paper is and what it is not	6
1.3 The intellectual lineage	6
1.4 Reading guide	7
2 The Theoretical Foundation: A Reference Recap	8
2.1 The formal setting	8
2.2 The three failure modes	8
2.3 The admissibility sources	9
2.4 The recovery mechanisms	9
2.5 The architecture bounds	11
3 The Quotient-Faithfulness Framework in the Landscape of Representation Learning Theory	11
3.1 Natural evolution: frameworks our work extends	11
3.2 Complementary instruments: frameworks that measure adjacent structure	13
3.3 Competing mechanisms: where predictions genuinely diverge	14
3.4 Different explanatory unit: the circuit challenge and its resolution	15
3.5 Three falsifiable predictions	15
4 Empirical Evidence: Fourteen Reframings	16
4.1 Group I: The three failure modes operating in practice	16
4.2 Group II: Recovery mechanisms working and failing	20
4.3 Group III: Scale and convergence	23
4.4 Group IV: The traversal-as-feature inversion	25
5 Open Problems as Reframing Problems	25
5.1 The certification gap: when can quotient faithfulness be verified?	25
5.2 Metric enforcement: when topology is not enough	26
5.3 Discovery upper bounds: from existence to learnability	26
6 What Would Falsify These Claims?	27
6.1 Framework-level falsifiers	27
6.2 The three distinguishing predictions	28
6.3 What confirmation would look like	29
A Definitions and Key Concepts	34
B Contribution Attribution and AI Disclosure	34
B.1 Context	34
B.2 Contribution table (reframing-relevant entries)	34
C Attribution Methodology	36

Notation

This paper uses the notation established in Papers I–III of the series. Key symbols are collected here for reference.

Symbol	Meaning
$\mathcal{X}$	Observation space
$\sim$	Admissible nuisance relation
$Q = \mathcal{X}/\sim$	Lawful quotient
$q : \mathcal{X} \rightarrow Q$	Canonical quotient projection
$E : \mathcal{X} \rightarrow Z$	Encoder
$\tilde{E} : Q \rightarrow Z$	Induced quotient representation
E_{label}	Label representation $E_{\text{label}}(x) = f(x)$
$E_1 \preceq E_2$	Compression preorder
$\text{emb}(Q)$	Minimum Euclidean embedding dimension of Q
$S(\theta)$	Batch-instability functional $W(\theta) + B(\theta)$
$\Sigma(Q, Z)$	Discriminant: boundary of C^1 embeddings $Q \hookrightarrow Z$
$K(x \mid n)$	Kolmogorov complexity (Paper 2, Remark 2.2a)

1 Introduction and Reading Guide

1.1 The thesis

A neural network that achieves perfect task accuracy, satisfies the minimal sufficient statistic criterion, and optimises any information-bottleneck objective has not necessarily learned anything about the equivalence structure of the process that generated its training data. Paper I of this series [Farkas, 2026a] proved that this gap is not an artefact of finite data or imperfect optimisation—it is structural. Three independent theorems establish that standard training criteria are blind to quotient structure: any finite sample admits a traversal solution achieving perfect loss; the minimal sufficient statistic collapses the semantic quotient whenever the task map is non-injective; and no topology-faithful encoding of a quotient Q exists in a space of dimension below $\text{emb}(Q)$. Papers 2 [Farkas, 2026b] and 3 [Farkas, 2026c] established what is needed to escape these traps and characterised four recovery mechanisms: stochasticity, structural regularisation, architecture design, and population selection.

What the six traditions examined here have in common is a proximity to the gap between *task-relative adequacy* and *lawful abstraction*: each tradition addresses part of what it means to learn a faithful representation, without making explicit which equivalence relation the representation is supposed to respect. The admissibility declaration $\sim$ is the structural variable that names that gap precisely; making it explicit is what the present series contributes.

This paper asks a different question. The quotient-faithfulness framework did not emerge from a vacuum. Six major theoretical traditions have been working on related problems. This paper maps where the quotient-faithfulness framework stands relative to each tradition, and then examines fourteen bodies of empirical evidence that become interpretively coherent once that map is in place.

The thesis is not that prior frameworks were wrong. Each identified something real and added something true. What the quotient-faithfulness framework contributes is a place to state explicitly which equivalence relation a representation is meant to respect—a structural component that each tradition typically leaves implicit.

Geometric Deep Learning handles the case where the symmetry group is known and fully specifies the admissibility relation by design. The Information Bottleneck provides a compression language but does not distinguish lawful from unlawful compression without an explicit admissibility declaration. Self-supervised anti-collapse methods give engineering solutions to representation collapse without a formal account of which distinctions collapse is structurally permitted to erase. Causal representation learning identifies the right target but leaves the obstacles to reaching it—the three failure modes—uncharacterised. Disentanglement theory identifies the right ambition and proves it ill-posed without inductive biases, without specifying what those biases should be. The flat-minima tradition identifies a relevant geometric phenomenon in parameter space without a parameterisation-invariant formalisation; the $S(\theta)$ batch-instability asymmetry provides one such formalisation, though not a resolution of the flat-minima identification problem in full generality.

The framework map in Section 3 makes these comparisons precise. The admissibility declaration $\sim$ is a missing structural component in these traditions—not the only thing they lack, and not a replacement for what they have established, but a formal articulation of a structural question each tradition leaves open.

1.2 What this paper is and what it is not

This paper is a perspective piece, not a primary research contribution. It proves no new results. Its contribution is conceptual: it situates the quotient-faithfulness framework precisely in the theoretical landscape of representation learning, shows that fourteen bodies of existing experimental evidence become interpretively coherent within that framework, and identifies the conditions under which the framework’s practical claims would require revision.

The fourteen reframings are illustrations of the framework map, not its primary content. A reader who finishes section 3—the framework map—before reading section 4—the reframings—will understand what the paper is arguing and why it matters.

Each reframing carries an evidence label reflecting the tightness of the connection. No reframing is presented as a derived consequence where the link is interpretive. No reframing refutes the original account it reframes. In every case, the original explanation remains correct within its own terms.

This paper makes four original contributions. First, it identifies the admissibility declaration $\sim$ as the specific structural ingredient missing from each of six prior traditions, and shows precisely where each tradition’s programme becomes underdetermined without it. Second, it provides the first mechanistic account of the Platonic Representation Hypothesis, explaining why scale and diversity weaken each of the three failure modes independently. Third, it resolves the apparent conflict between mechanistic interpretability and quotient faithfulness by showing that the circuit account and the quotient account operate at different levels of description. Fourth, it generates three falsifiable predictions that distinguish the framework from specific competing accounts and have not been tested by any published study.

1.3 The intellectual lineage

The paper has two chains of intellectual descent.

Chain 1: The invariance and compression tradition. Geometric Deep Learning [Bronstein et al., 2021] is the seed: it identified group-equivariant architecture as the right inductive bias when the symmetry group is known, raising the obvious question of what happens when it is not. The Information Bottleneck [Tishby et al., 1999, Tishby and Zaslavsky, 2015] provided the compression language. Self-supervised anti-collapse methods [Chen et al., 2020, Grill et al., 2020, Zbontar et al., 2021, Bardes et al., 2022, LeCun, 2022] operationalised the compression problem for self-supervised learning. Causal representation learning [Schölkopf et al., 2021] identified the causally stable variables as the right target. Locatello et al. [2019] proved that the target is unidentifiable without explicit inductive biases. Our framework provides what Locatello showed is necessary: the admissibility declaration $\sim$.

Chain 2: The optimisation geometry tradition. Hochreiter and Schmidhuber [1997] identified the flat-minima intuition: solutions in flat loss basins generalise better. This was the independent geometric starting point for the batch-instability asymmetry. Dinh et al. [2017] showed the intuition could not be formalised through the Hessian. SAM [Foret et al., 2021] operationalised the intuition in practice without resolving the theoretical problem. The $S(\theta) = W(\theta) + B(\theta)$ decomposition (Propositions 5.6A, 5.6C; Lemma 5.6D in Paper 2 [Farkas, 2026b]) is the parameterisation-invariant formalisation: gradient scatter across mini-batches is a property of the loss landscape relative to the data, not of the weight coordinates.

1.4 Reading guide

section 2 recaps the theoretical foundation of Papers 1, 2a, and 2b without restating proofs. A reader who has read all three prior papers can skip it and treat it as a reference.

section 3 is the paper’s centrepiece. It maps the quotient-faithfulness framework against six theoretical traditions in three tiers: natural evolution (section 3.1), complementary instruments (section 3.2), and competing mechanisms (section 3.3). A fourth subsection addresses the mechanistic interpretability challenge and its resolution (section 3.4). The section closes with three falsifiable predictions (section 3.5).

section 4 presents the fourteen reframings as illustrations of section 3’s claims. The reframings are grouped by what they illustrate: the three failure modes operating in practice (section 4.1), recovery mechanisms working and failing (section 4.2), scale and convergence (section 4.3), and the traversal-as-feature inversion (section 4.4).

section 5 identifies three open problems that emerge from the gaps the framework map reveals.

section 6 states the conditions under which each major practical claim of the framework would require revision.

Section	Prior papers needed
§2 recap	None—§2 is self-contained
§3.1 (GDL, IB, SSL, causal RL, disentanglement)	Paper I §1–2 (three failure modes); Paper 2 §3 (admissibility sources)
§3.2 (SLT, flat minima, MDL, MoE)	Paper 2 §5 ($S(\theta)$); Paper 3 §3 (dimension floor)
§3.3 (implicit bias, NTK, benign overfitting, PAC-Bayes)	Paper I §3–4; Paper 2 §5 (batch-instability)
§3.4 (mechanistic interpretability)	Paper 2 §5.6P (discriminant crossing)
§4 reframings 1–5 (three failure modes)	Paper I (all three failure modes)
§4 reframings 6–9 (recovery)	Paper 2 §5 (stochasticity), §3 (admissibility)
§4 reframings 10–12 (scale & convergence)	Paper 3 §3 (dimension floor); Paper 2 §3.4 (graded admissibility)
§4 reframings 13–14 (inversion)	Paper 2 §5.6P; Paper 3 §3
§5 open problems	Papers 2 §9 and 3 §12
§6 falsifiability	Paper 2 §5 (Propositions 5.6A, 5.6C, 5.6E)

Table 1: Reading guide: which prior papers are needed for each section.

What this paper does not claim. The reframings in §§4.1–4.3 are missing-variable analyses, not explanations. Each one identifies the admissibility declaration $\sim$ as a structural component that the cited tradition leaves implicit. None of them claim that the quotient-faithfulness framework *subsumes* or *explains* the cited tradition at its own level of description. IB has a complete theory of compression; this series adds the structural variable that distinguishes lawful from unlawful compression. GDL has a complete theory of equivariance; this series characterises the case when the symmetry group is not declared. Mechanistic interpretability has a complete account of circuits; this series asks what structural property the circuit must realise. The framework is strongest as a diagnostic instrument for what is missing, not as a unified theory of what

exists.

2 The Theoretical Foundation: A Reference Recap

This section recaps the results from Papers 1, 2a, and 2b that Paper 5 builds on. Each result is cited by paper and theorem number and labelled by epistemic status: **[Proved]**, **[Conditional]**, **[Rigorous local model]**, or **[Conjectural]**. A reader who has read all three prior papers can skip this section and treat it as a reference when later sections cite specific results.

2.1 The formal setting

A *data-generating process* produces observations $x \in \mathcal{X}$ from an underlying state space with a declared *nuisance relation* $\sim$ on $\mathcal{X}$: $x \sim x'$ means the two observations differ only in variation declared admissible. The *lawful quotient* is $Q = \mathcal{X}/\sim$, the space of semantically distinct states. A *representation* is a measurable encoder $E : \mathcal{X} \rightarrow Z$ where $Z = \mathbb{R}^m$ is the exposed latent space.

Three properties together constitute *quotient faithfulness* of E :

- (I) **Sufficiency**: E retains enough information to predict the task target Y .
- (II) **Quotient-respecting**: $E(x) = E(x')$ whenever $x \sim x'$.
- (III) **Topology-faithful**: The induced map $\tilde{E} : Q \rightarrow Z$ is a topological embedding.

Properties (I)–(III) are logically independent. The *task map* $t : Q \rightarrow Y$ maps semantic states to prediction targets. The *label representation* $E_{\text{label}} = t \circ q$ satisfies (I) and (II) while failing (III) whenever t is non-injective.

2.2 The three failure modes

Result (Paper I, Theorem 4.6 — Finite-Sample Traversal). *[Proved]*

For any finite training set $D = \{(x_i, y_i)\}_{i=1}^N$, there exists a continuous injective curve $\gamma : [0, 1] \rightarrow \mathcal{X}$ passing through every sample point. A one-dimensional traversal coordinate along γ achieves perfect samplewise task loss. The samplewise objective cannot distinguish this traversal from a quotient-faithful representation. (In Paper I’s revised framing, this is the $k = 1$ case of graded compression; Theorem 4.7 gives the full graded family.)

What this means. Finite task performance provides no evidence that quotient structure has been recovered. The result concerns the loss landscape, not training dynamics.

Result (Paper I, Theorem 4.4 — Task-Map Collapse). *[Proved]*

The global minimum of the compression objective is $E_{\text{label}} = t \circ q$. Whenever t is non-injective on Q , E_{label} collapses Q : it maps distinct semantic states to the same code. This minimum exists in the population risk. No task-based criterion can detect this collapse because E_{label} is optimal by the criterion’s own measure.

What this means. The loss landscape has its global minimum at a structurally wrong point when t is non-injective. Escaping requires a mechanism that reshapes the landscape—stochasticity alone is insufficient because the minimum is population-

level.

Result (Paper I, Corollary 3.6 — Below-Floor Collapse). *[Proved]*

Let Q be a compact metric space with covering dimension k . If $\dim(Z) < k$, no continuous map $\tilde{E} : Q \rightarrow Z$ can be topology-faithful. This applies to the final encoder output Z regardless of internal architecture, layer widths, skip connections, or training procedure.

What this means. The dimension of the exposed latent space is a hard floor on what faithful encodings are possible, independent of learning.

2.3 The admissibility sources

The framework identifies four currently characterised sources by which $\sim$ enters the learning process, in decreasing order of evidential strength [Farkas, 2026b, Paper 2, §3]:

1. **Known group action** [Full certification]: Equivariant architectures enforce $E(g \cdot x) = \rho(g) \cdot E(x)$ by construction. Condition (II) is satisfied as an architectural identity.
2. **Causal invariances** [Near-certification when coverage is complete]: $x \sim x'$ iff they share the same intervention-stable content under causal interventions on the data-generating process.
3. **Augmentation and multiview supervision** [Strong evidence, misspecification risk]: Contrastive objectives treat augmentation-equivalent pairs as nuisance-equivalent. Misspecification invalidates the certification claim.
4. **Domain declarations and scale-convergence** [Weak signal]: Task diversity, data diversity, and scale provide evidence that something like a shared quotient structure exists, but do not certify which one.

Result (Paper 2, Theorem 3.5 — Graded Admissibility). *[Proved]*

The hard binary relation $\sim$ generalises to a graded quotient kernel $r : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$, equivalent to a semantic ultrapseudometric d_c on $\mathcal{X}$ and to a nested filtration of hard quotients $\{Q_{c,\varepsilon}\}$ indexed by semantic scale ε .

Result (Paper 2, Theorem 3.7 — Scale-by-Scale Certification). *[Proved]*

If the semantic ultrapseudometric d_c can be verified at scale ε , then the hard sub-quotient $Q_{c,\varepsilon} = \mathcal{X} / \sim_{c,\varepsilon}$ is certified as represented in Z up to explicitly bounded distortions.

2.4 The recovery mechanisms

The three failure modes are remediable to different degrees through four mechanisms [Farkas, 2026b, Paper 2, §5]:

Stochasticity [Traversal trap only]:

Result (Paper 2, Proposition 5.6A — Exact Batch-Instability Scaling). *[Proved]*

$\mathcal{V}(\theta) = \frac{N-b}{b(N-1)} \cdot S(\theta)$ exactly, where $S(\theta) = W(\theta) + B(\theta)$ decomposes gradient scatter into within-class nuisance sensitivity $W(\theta)$ and between-class gradient dispersion $B(\theta)$.

Result (Paper 2, Proposition 5.6C — Traversal Instability Bound). *[Conditional]*

Under named coverage and nuisance-variability conditions, $S(\theta_{\text{trav}}) \geq \delta_W^2 + \delta_B^2 > 0$. Traversal solutions have measurably higher gradient scatter than quotient-aligned solutions under these conditions.

Result (Paper 2, Lemma 5.6D — Local Escape Probability). *[Rigorous local model]*

The one-step escape probability from a traversal tube is $\approx 2\Phi(-r/\eta\sigma_u)$ where $\sigma_u^2 = u^\top \Sigma(\theta_\star)u$. Escape probability is increasing in $S(\theta)$ and decreasing in batch size b .

Result (Paper 2, Proposition 5.6P — Discriminant Crossing is Necessary). *[Proved]*

Any continuous training trajectory from a traversal-type solution to a quotient-faithful embedding must cross the discriminant $\Sigma(Q, Z)$ —the boundary of the space of C^1 embeddings. This is a necessary topological condition on any escape path, independent of mechanism.

Conjecture (Paper 2, Conjecture 5.6E — Directed Statistical Eviction). *[Genuinely conjectural]*

Escape from traversal solutions is directed toward quotient-aligned solutions—batch-conditioned curvature failure concentrates displacement in the transverse direction. This is the single unproved step in the stochasticity account. Falsifiable from training traces by checking whether escape events correlate with mini-batches where the batch-conditioned projected curvature is non-positive.

Result (Paper 2, Corollary 5.8.6 — No Universal Recovery Mechanism). *[Proved]*

No single mechanism escapes all three failure modes. Stochasticity escapes the finite-sample traversal but not the graded compression, because E_{label} is the global minimum of the population risk and no single-network noise mechanism changes that minimum’s location.

Structural regularisation [Compression trap]: Topology losses, metric constraints, equivariant architectures, and JEPA-style noise raise the effective energy of the graded-compression minimum relative to quotient-faithful alternatives. None guarantees escape.

Architecture design [Dimension trap]: The only remedy for the embedding floor is ensuring $\dim(Z) \geq \text{emb}(Q)$.

Population selection [Compression trap, fourth mechanism]: Selection over a population under a fitness criterion that includes structural measures beyond task loss can escape the graded compression by changing the effective criterion.

2.5 The architecture bounds

Result (Paper 3, Proposition 3.1 — Width Floor). *[Proved]*

Any topology-faithful encoding $\tilde{E} : Q \rightarrow Z$ requires $\dim(Z) \geq \text{emb}(Q)$. This constrains the final encoder output Z , not any internal layer. A non-bypassable layer (one through which every directed path passes with no bypass) must also have width $\geq \text{emb}(Q)$. Modern architectures with skip connections, attention, or MoE routing generally do not contain non-bypassable internal layers.

Result (Paper 3, Corollary 3.2 — Product Decomposition). *[Proved]*

For $Q = Q_1 \times \cdots \times Q_k$, $\text{emb}(Q) = 1 + \sum_i \dim(Q_i)$ under the sphere-product formula.
For $Q = (S^1)^n$ (n -torus): $\text{emb}(Q) = n + 1$.

The embedding hierarchy for smooth manifolds: Menger–Nöbeling ($\text{emb}(Q) \leq 2n + 1$), Whitney ($\leq 2n$), and Haefliger ($\leq 2n - k$ for k -connected quotients, $n \geq 2k + 3$). The upper bound programme of Paper 3 answers existence, not learnability—whether gradient descent finds a faithful encoding at the minimum sufficient width is the discovery upper bounds open problem (section 5.3).

3 The Quotient-Faithfulness Framework in the Landscape of Representation Learning Theory

Every framework in this map is attempting to answer some version of the same question: what makes a learned representation good? They differ in what “good” means, what structure they assume is available, and what failure modes they can see. The synthesis below maps where each framework stands on these three dimensions and shows how the quotient-faithfulness framework relates to each.

3.1 Natural evolution: frameworks our work extends

Geometric Deep Learning — the seed, and the gap it left. [Bronstein et al. \[2021\]](#) synthesised a powerful organising principle: the right inductive bias for a learning problem is the symmetry group of the data-generating process, built into the architecture by construction. When the nuisance structure is a known compact Lie group G acting on $\mathcal{X}$, equivariant architectures enforce $E(g \cdot x) = \rho(g) \cdot E(x)$ by construction—the group-action admissibility source of Paper 2 §3.1. The architecture IS the admissibility declaration.

The gap GDL left is the boundary of its own assumptions: it requires G to be known and correctly specified at design time. A network equivariant under $\text{SO}(2)$ correctly collapses in-plane image rotations—but fails on out-of-plane tilts, scale variation, and illumination change, unless those are also explicitly declared. GDL’s formalism has no tools for this misspecification case. The admissibility declaration is the missing structural variable: GDL solves the admissibility problem completely when the symmetry group is known and correctly specified; what it leaves implicit is how to proceed when the group is unknown, misspecified, or only partially evidenced from data.

Information Bottleneck — the compression language, and what $\sim$ adds. The IB [[Tishby et al., 1999](#), [Tishby and Zaslavsky, 2015](#)] frames representation learning as optimisation over mutual information: find the representation Z that retains maximum

information about target Y while discarding maximum information about input X . The IB Lagrangian is $\min I(X; Z) - \beta \cdot I(Z; Y)$.

This is the right language. But when the task map t is non-injective on Q , the IB optimal representation is $E_{\text{label}} = t \circ q$ —the graded-compression minimum of Theorem 4.4. IB names it as the optimal target; we name it as the structural failure. The same mathematical object— E_{label} —is the goal of IB and the trap of our framework.

The missing structural component is $\sim$: the declared admissibility relation that specifies which compressions are lawful (collapsing nuisance variation within fibers of $\sim$) and which are not (collapsing Q 's structure in fibers of t). The missing structural variable is $\sim$: without it, IB cannot distinguish compressions that are lawful (collapsing nuisance variation within fibres of $\sim$) from compressions that are not (collapsing Q 's structure in fibres of t). The series does not extend IB mathematically; it names the gap in what IB's objective selects for.

Self-supervised anti-collapse — the engineering solutions, and what theory they were missing. The SSL anti-collapse family—SimCLR [Chen et al., 2020], BYOL [Grill et al., 2020], Barlow Twins [Zbontar et al., 2021], VICReg [Bardes et al., 2022], JEPA [LeCun, 2022]—emerged from a practical crisis: joint-embedding methods collapse. The trivial solution, outputting the same constant vector for every input, achieves perfect invariance while learning nothing. Every method in this family is an engineering solution to collapse prevention.

Collapse is the graded compression in its most extreme form. But the SSL literature conflates two distinct failure modes: *dimensional collapse* (representations lose variance or rank) and *semantic collapse* (distinct elements of Q map to the same code even though the codes are not identical and not constant). Anti-collapse mechanisms prevent the former; they say nothing about the latter. A Barlow Twins representation can have full-rank covariance while still encoding E_{label} rather than a faithful representation of Q . The missing structural variable is the specification of which distinctions may be collapsed: anti-collapse mechanisms address dimensional collapse without addressing semantic collapse, because the admissibility declaration $\sim$ that separates lawful from unlawful collapse is never made explicit in the SSL objective.

Causal representation learning — the target specification, and the obstacle characterisation it lacked. Schölkopf et al. [2021] identified the right target: the causally stable variables, those remaining invariant under interventions on the data-generating process. Richens and Everitt [2024] proved formally that agents generalising robustly under arbitrary interventions must have learned the causal graph. The causal variables are what should be encoded; everything else is nuisance.

This is our admissibility source §3.1.2 stated in causal language. The quotient $Q = \mathcal{X}/\sim$ is the space of causally stable content. Invariant Risk Minimisation [Arjovsky et al., 2019] operationalises this through multi-environment training; its documented failure modes [Rosenfeld et al., 2021, Gulrajani and Lopez-Paz, 2021] are traversal instances— incomplete environment coverage means a traversal of the available environments satisfies IRM's criterion without encoding the true causal quotient.

What causal representation learning was missing: the three structural traps that prevent standard training from reaching a faithful encoding even when Q is correctly specified. Our framework adds the obstacle characterisation.

Disentangled representation learning — the ambition, and the achievable target. Disentanglement [Bengio et al., 2013, Locatello et al., 2019] asks for representations where each latent dimension independently encodes one factor of variation. This is a strictly stronger requirement than quotient faithfulness: we require that Q be faithfully encoded; disentanglement requires that every factor—including nuisance factors—be independently encoded.

Locatello et al. [2019] proved that without inductive biases, disentanglement is unidentifiable from data alone. This is our certification problem at a higher level of strictness. Locatello’s result motivates asking what weaker targets remain achievable with explicit inductive bias. The admissibility declaration $\sim$ is one precise such bias: it makes quotient faithfulness a well-posed target without claiming to supply the complete inductive structure that full disentanglement would require. The three-trap structure then characterises when even that weaker target eludes standard training.

3.2 Complementary instruments: frameworks that measure adjacent structure

Singular Learning Theory and the grokking transition. SLT [Watanabe, 2009] characterises loss basin geometry through the Real Log Canonical Threshold (RLCT) λ . Smaller λ means better generalisation at finite sample size. The grokking transition in modular arithmetic corresponds to a decrease in RLCT—the network moves from a high-RLCT traversal minimum to a low-RLCT quotient-aligned minimum.

The developmental interpretability programme [Hoogland et al., 2023] uses RLCT to track phase transitions during training. Our $S(\theta)$ asymmetry and the RLCT dynamics are both tracking the traversal-to-faithful transition through different instruments. SLT gives the geometry; our framework gives the semantics.

Flat minima and the $S(\theta)$ formalisation. The flat-minima intuition [Hochreiter and Schmidhuber, 1997] identifies the right phenomenon—stable solutions generalise—but cannot be formalised through the Hessian because sharpness is parameterisation-dependent [Dinh et al., 2017]. SAM [Foret et al., 2021] operationalises the intuition without resolving the theoretical problem.

$S(\theta) = W(\theta) + B(\theta)$ is the rigorous, parameterisation-invariant formalisation: gradient scatter across mini-batches is a property of the loss landscape relative to the data distribution, not of the weight coordinates. Reparameterising the network changes the Hessian but not $S(\theta)$. SAM can be interpreted as implicitly favouring low- $S(\theta)$ solutions: its worst-case perturbation step pushes away from high-gradient-scatter configurations, which correspond to traversal solutions. This interpretation is a structural parallel, not a proved equivalence; the batch-size scaling prediction of Proposition 5.6A would help discriminate it empirically.

Rate-distortion and the topology floor. Rate-distortion theory [Shannon, 1959] establishes the information-theoretic compression floor: no representation can achieve distortion below D with fewer than $R(D)$ bits. Our Corollary 3.6 establishes the topological floor: no faithful encoding of Q exists in a space of dimension below $\text{emb}(Q)$. Both constraints are binding simultaneously. Corollary 5.7.3 in Paper 2 is the explicit bridge: the NSW threshold connects the noise level to the reach τ of Q and the Jacobian spectrum.

Mixture of Experts as local quotient covering. MoE architectures [Shazeer et al., 2017, Fedus et al., 2022] implement a natural covering of Q by expert-specialised approximations. Each expert learns a locally faithful representation of a patch of Q ; the gating function implements the transition between patches—the structure of a manifold atlas. Expert collapse is the MoE analogue of the finite-sample traversal at the routing level; load balancing losses are admissibility enforcement at the routing level.

Kolmogorov complexity and simplicity bias. Dingle and Hutter [2026] prove that under simple objectives over simple solution spaces, optima must have low Kolmogorov complexity, and that simple optima of different objectives are exponentially more likely to coincide than a random-objective null model predicts. Deep networks implement exactly this prior [Valle-Pérez et al., 2018, Mingard et al., 2025]. This constitutes a seventh adjacent tradition not captured in the six above: it operates at the level of a priori selection rather than convergence geometry or certification.

Framework-map assignment: what it contributes is a principled reason why KC-simple solutions are preferentially found, and a coincidences result explaining Platonic convergence a priori. What it does not resolve is which KC-simple solution is semantically correct: E_{label} and quotient-faithful encoders of simple Q are both KC-simple; simplicity bias cannot distinguish them without an admissibility declaration. The combination: Dingle–Hutter explains why certain structures are a priori more likely; the quotient-faithfulness framework explains why likelihood is not the same as semantic correctness. The admissibility declaration is the structural component that breaks the tie.

3.3 Competing mechanisms: where predictions genuinely diverge

The implicit bias family — aligned on the finite-sample traversal, opposed on the graded compression. Implicit bias results [Lyu and Li, 2020, Ji and Telgarsky, 2019, Gunasekar et al., 2017] establish that gradient descent selects among interpolating solutions according to complexity criteria implicit in the optimiser and architecture. For homogeneous networks with exponential-tailed losses, gradient flow converges to the maximum-margin classifier.

On the finite-sample traversal, the frameworks are aligned: high-norm traversal solutions are disfavoured by implicit bias toward low-norm solutions. Our $S(\theta)$ asymmetry gives the structural reason—traversal solutions are sample-specific and thus high-norm relative to quotient-aligned solutions.

On the graded compression, the frameworks are opposed: the max-margin classifier under a classification loss is exactly $E_{\text{label}} = t \circ q$. Implicit bias toward max-margin drives toward the graded compression, not away from it.

The distinguishing prediction: Does grokking occur without weight decay? Our batch-instability account (the $S(\theta)$ mechanism operating independently of norm regularisation) predicts yes. The norm-penalty account predicts no. Evidence from Thilak et al. [2022] via the slingshot mechanism supports our account; a 2025 paper [Barak et al., 2025] shows grokking without weight decay by preventing Softmax Collapse—an independent numerical mechanism. Both the SC and $S(\theta)$ accounts may operate simultaneously; the batch-size scaling prediction (Proposition 5.6A) specifically distinguishes our account.

NTK and lazy training — the kernel regime is the traversal-dominant regime. The NTK framework [Jacot et al., 2018] establishes that in the lazy training regime, the unique solution is the minimum-norm kernel interpolant of the training set—a traver-

sal solution in our vocabulary. The rich-to-lazy transition is the traversal-to-faithful transition described at the optimisation-theoretic level.

The distinguishing prediction: Networks trained in the kernel regime should show lower quotient-faithfulness than rich-regime networks at matched task accuracy.

Benign overfitting — task accuracy is independent of quotient faithfulness. Benign overfitting [Bartlett et al., 2020, Belkin et al., 2019] establishes that interpolating models can generalise well on task loss. This is compatible with our framework but concerns a different question: task-loss generalisation, not quotient faithfulness.

The distinguishing prediction: Models satisfying benign overfitting conditions can simultaneously fail quotient-faithfulness tests when t is non-injective on Q —as demonstrated by the mirage reasoning finding (section 4.1).

PAC-Bayes — orthogonal complexity measure, compatible in the overparameterised regime. PAC-Bayes [McAllester, 1999, Dziugaite and Roy, 2017] bounds generalisation gap through posterior-prior KL divergence. In the underparameterised regime, a traversal solution found near initialisation can have low KL divergence while being structurally wrong. In the overparameterised regime, the prior’s broad support covers quotient-aligned solutions, and low posterior-prior KL tracks solution stability—which tracks quotient alignment. The residual divergence: PAC-Bayes controls task-loss generalisation and cannot detect semantic collapse.

3.4 Different explanatory unit: the circuit challenge and its resolution

Mechanistic interpretability [Elhage et al., 2021, Olsson et al., 2022, Nanda et al., 2023] explains model behaviour through circuits—specific computational subgraphs implementing specific algorithms. The explanatory unit is an internal algorithm, not a representation geometry.

The apparent conflict with our framework resolves when examined precisely. Chughtai et al. [2023] show that networks trained on group composition learn the irreducible representations of the group as their internal encoding. For the cyclic group $\mathbb{Z}/p\mathbb{Z}$, the irreducible representations are exactly the Fourier modes that Nanda et al. [2023] found in modular arithmetic grokking. The circuit account and the quotient-faithfulness account are not competing alternatives: the circuit IS the mechanism by which quotient faithfulness is achieved.

Complementarity is precise: circuits explain *how* a specific model implements a specific computation; our framework identifies *what structural property* the computation must achieve and *why standard training struggles* to achieve it. A key clarification: our framework does not require quotient faithfulness to be encoded in any single layer. The faithful encoding $\tilde{E} : Q \rightarrow Z$ is a property of the full computational map from input to representation, however that map is distributed across layers, heads, and MLP blocks.

3.5 Three falsifiable predictions

Three predictions where the frameworks make experimentally separable claims:

Prediction 1 (Grokking mechanism). The $S(\theta)$ batch-instability account predicts that grokking transition timing scales with batch size as $(N - b)/b$ (Proposition 5.6A). The norm-penalty account predicts weight decay is necessary and transition timing is

independent of batch size. These are distinguishable by tracking $S(\theta)$ decomposition and batch-size dependence of transition timing in a controlled grokking experiment.

Prediction 2 (Kernel-regime quotient faithfulness). Networks trained in the lazy / NTK regime should show lower quotient-faithfulness than rich-regime networks at matched task accuracy. Testable using tasks with known Q (group composition, Othello move prediction) and mechanistic probes for quotient structure.

Prediction 3 (Task accuracy independence). Task accuracy and quotient faithfulness are independent: high accuracy is achievable with zero faithfulness (mirage reasoning) and faithfulness does not require high accuracy on any particular task. Testable by cross-comparing task accuracy and quotient-faithfulness scores across architectures and training conditions.

4 Empirical Evidence: Fourteen Reframings

The fourteen reframings below are illustrations of the framework map developed in section 3, not the paper’s primary claim. Each reframing is assigned an evidence label from a five-tier scale (Table 2). Each states the original explanation faithfully, identifies where our framing aligns and where it adds, and states explicitly what it does not claim.

Label	Meaning
[Direct structural evidence]	Reframing follows from proved results; empirical finding matches a quantitative prediction
[Strong compatibility]	Design intent or documented failure mode maps directly onto the framework without additional interpretive steps
[Compatible observation]	Consistent with the framework; alternative explanations exist; identification requires interpretive steps
[Structural parallel; mechanism differs]	Structural relationship identified; original mechanism differs from ours

Table 2: Evidence label definitions.

4.1 Group I: The three failure modes operating in practice

Reframing 1. mLaSDI’s competing-objectives problem as the graded compression in scientific ML. [Direct structural evidence]

Original explanation. The LaSDI family [Tran et al., 2023, Anderson et al., 2025] trains autoencoders on high-dimensional PDE simulation data with two simultaneous objectives: accurate reconstruction, and ODE-consistent latent dynamics. Anderson et al. explicitly document a competing-objectives problem and resolve it by staging training—dynamics-respecting structure is learned before reconstruction is enforced.

Alignment. The chain is tight. The ODE structure is the admissibility source: it declares which variations in the high-dimensional state space are dynamically equivalent, defining the target quotient Q . The reconstruction objective is the task map t : non-injective on Q whenever dynamically distinct states produce similar reconstructed outputs. By Theorem 4.4, the global minimum of the reconstruction objective is E_{label} —a representation that collapses the dynamical distinctions the reconstruction cannot see. The competing-objectives problem is the graded compression operating exactly as the theorem predicts.

#	Phenomenon	Label	Trap / mechanism	Key sources
<i>Group I: The three failure modes operating</i>				
1	mLaSDI competing objectives	Direct structural	Compression trap	Anderson et al. [2025]
2	Mirage reasoning	Direct structural	Traversal trap at scale	Asadi et al. [2026]
3	Grokking	Three proved + one conj.	Traversal-to-faithful	Power et al. [2022], Nanda et al. [2023], Chughtai et al. [2023]
4	IRM failure modes	Strong compatibility	Traversal trap under causal admissibility	Arjovsky et al. [2019], Rosenfeld et al. [2021]
5	Neural collapse	Compatible observation	Compression trap geometry	Papayan et al. [2020]
<i>Group II: Recovery mechanisms</i>				
6	JEPA	Strong compatibility	Compression-trap escape	LeCun [2022], Garrido et al. [2026]
7	Transversal ARC	Direct structural	Structure-first recovery	Khalil [2026]
8	Emergent world models	Strong compatibility	Spontaneous quotient discovery	Li et al. [2023], Gurnee and Tegmark [2024]
9	Score-based diffusion models	Strong structural parallel	Systematic graded quotient discovery	Song et al. [2021], Ho et al. [2020]
<i>Group III: Scale and convergence</i>				
10	Platonic Representation Hypothesis	Strong compatibility	All three failure modes weakened	Huh et al. [2024]
11	Superposition	Compatible observation	Dimension trap (inverse)	Elhage et al. [2022]
12	Mode connectivity	Compatible observation	Parameter-space heterogeneity	Draxler et al. [2018], Liao et al. [2024]
<i>Group IV: Traversal-as-feature inversion</i>				
13	Speculative and speculative-speculative decoding	Structural parallel	Quotient traversal with faithful verification	Leviathan et al. [2023], Kumar et al. [2026]
14	Lottery tickets	Compatible observation	Distributed fragility	Frankle and Carlin [2019]

Table 3: Summary of the fourteen reframings.

Addition. The staged training solution is a specific instance of the general principle: structural regularisation must reshape the loss landscape before the task loss drives the representation into the graded compression.

What we do not claim. We do not claim the graded compression is the only source of difficulty in LaSDI training. Numerical stability, ODE integrator choice, and dataset

coverage all play roles our framework does not characterise.

Reframing 2. Mirage reasoning as traversal-consistent multimodal collapse. [Direct structural evidence]

Original explanation. Asadi et al. [2026] document that frontier multimodal models generate detailed image descriptions and achieve high benchmark scores with no image input at all. In the most extreme case, a model attained top rank on a chest X-ray question-answering benchmark without accessing any images.

Alignment. This is Theorem 4.6 operating at deployment scale. The task map t is the benchmark evaluation criterion—it is non-injective on the visual quotient Q_{visual} because language priors alone constitute a finite-sample fit achieving performance indistinguishable from genuine visual encoding. No recovery mechanism has enforced encoding of Q_{visual} . The structural match is the tightest of any reframing: the impossibility result says task performance cannot certify quotient faithfulness, and the empirical finding confirms this impossibility at the hardest possible scale.

Addition. Provides the benchmark design prescription: evaluation criteria for multimodal systems must test the structure of Q_{visual} , not just performance on finite evaluation sets.

What we do not claim. We do not claim that all multimodal models are operating in this regime.

Reframing 3. Grokking as traversal-to-faithful transition via statistical eviction. [Three proved results, one conditional bound, one conjectural step]

Original explanation. Power et al. [2022] reported delayed generalisation on algorithmic tasks. Nanda et al. [2023] identified the mechanism for modular addition: early training finds a memorisation (traversal) solution; continued training under weight decay builds Fourier features that implement the modular arithmetic algebraically. Chughtai et al. [2023] showed these Fourier features are irreducible representations of the cyclic group—the algebraic structure the task requires.

Alignment. The memorisation solution is a traversal threading the specific training pairs without encoding the algebraic structure of $\mathbb{Z}/p\mathbb{Z}$. The generalising solution is a faithful encoding of the cyclic quotient $Q = \mathbb{Z}/p\mathbb{Z}$, implemented through its irreducible representations. The transition is traversal-to-faithful. Proposition 5.6P establishes that this transition must cross the discriminant $\Sigma(Q, Z)$. Propositions 5.6A and 5.6C establish that traversal solutions have higher gradient scatter $S(\theta)$, providing the escape mechanism.

The circuit and quotient accounts are not competing. The Fourier features that Nanda et al. identified are the irreducible representations of $\mathbb{Z}/p\mathbb{Z}$ —the same objects our framework predicts should be learned. The circuit account names the mechanism; our framework names the structural property those features instantiate.

Addition. Explains why the memorisation phase exists (Theorem 4.6), why the transition is sharp (Proposition 5.6P), and provides the batch-size scaling prediction that distinguishes our account from the norm-penalty account (see section 3.5).

What we do not claim. Conjecture 5.6E—that escape is directed rather than diffusive—remains unproved. The reframing rests on four proved results and one conjectural step, and the evidence labels reflect this asymmetry.

Topological sector structure. Proposition 5.6P implies a sharper framing. Memorisation

and algebraic solutions lie in different topological sectors: the traversal solution has winding number zero on the cyclic quotient $\mathbb{Z}/p\mathbb{Z}$; the algebraic solution occupies the correct topological sector. In field-theory language, grokking is a *topological sector crossing*: the dynamics must cross $\Sigma(Q, Z)$ to reach the faithful solution, and this crossing is non-perturbative—no infinitesimal gradient step from the memorisation minimum reaches the algebraic minimum. The sharp transition is the signal of this crossing. The Fourier features of Nanda et al. [2023] are the soliton signature: irreducible representations of $\mathbb{Z}/p\mathbb{Z}$ carrying the topological charge of the correct sector. This framing is formalised as Open Problem 7 in Paper 2 [Farkas, 2026b].

Reframing 4. IRM failure modes as finite-sample traversal under causal admissibility. [Strong compatibility]

Original explanation. IRM [Arjovsky et al., 2019] trains a representation such that the optimal classifier on top of it is simultaneously optimal across all training environments. Documented failure modes [Rosenfeld et al., 2021, Gulrajani and Lopez-Paz, 2021] show IRM convergences to non-causal representations when training environments do not sufficiently span the space of spurious correlations.

Alignment. IRM’s failure mode is Theorem 4.6 in multi-environment form. For any finite collection of training environments, a traversal solution exists that satisfies IRM’s invariance criterion on training environments without encoding the true causal quotient Q . Incomplete coverage is incomplete admissibility specification.

Addition. The coverage conditions in Proposition 5.6C translate directly into conditions on environment diversity that IRM training requires. The finite-sample traversal provides the structural diagnosis that IRM’s own literature acknowledges without fully explaining.

What we do not claim. Computational intractability of the IRM objective, sensitivity to the invariance penalty, and architectural limitations all play roles our framework does not characterise.

Reframing 5. Neural collapse as graded-compression signature at scale. [Compatible observation]

Original explanation. Pappan et al. [2020] documented four geometric phenomena occurring simultaneously in the terminal phase of training: within-class variability collapses to zero (NC1), class means converge to a maximally-spread equiangular tight frame (NC2), classifier weights converge to the same ETF (NC3), and each input is classified by nearest class mean (NC4).

Alignment. NC1 is the graded compression instantiated at the classification scale. The task map t is the class label function, non-injective on Q whenever semantically distinct within-class inputs exist. The graded compression minimum E_{label} collapses all within-class variation to a single point—exactly NC1. NC2–NC4 describe the specific geometry of E_{label} under cross-entropy loss with balanced classes.

Addition. Explains why NC1 is structurally significant rather than merely a training artefact, and connects the ETF geometry to the global minimum of the compression objective.

What we do not claim. Maximum margin geometry and unconstrained features provide alternative accounts of the ETF geometry that are equally valid descriptions at a different level of analysis. Our framework and the geometric accounts are complementary.

4.2 Group II: Recovery mechanisms working and failing

Reframing 6. JEPA as a graded-compression architectural response. [Strong compatibility]

Original explanation. LeCun [2022] proposed JEPA as an alternative to generative and discriminative self-supervised methods, adding Gaussian noise to the latent representation during training. The noise prevents the predictor from collapsing to a trivial constant solution.

Alignment. The JEPA noise term is a direct architectural response to Theorem 4.4. A collapsed representation will have its collapsed codes spread apart by the noise, raising the effective energy of the graded-compression minimum relative to less-collapsed alternatives. The design intent maps directly onto our framework.

Addition. The noise term is a soft penalty—it raises the energy of the wrong minimum without removing it. Theorem 4.4 predicts that a hard structural regulariser is needed for full escape. JEPA will improve representations relative to no noise but will not fully escape the graded compression for non-injective task maps.

What we do not claim. JEPA does not specify which admissibility source supplies the target equivalence relation.

Reframing 7. Transversal ARC solving as structure-first quotient recovery. [Direct structural evidence]

Original explanation. Khalil [2026] solves ARC grid-transformation tasks with zero gradient descent by declaring projective geometric structure as the complete admissibility source and operationalising it via Schubert calculus transversals, achieving 86–95% success on non-timeout tasks.

Alignment. This is the paper’s positive prescription in its purest form. Complete admissibility declaration eliminates all three traps simultaneously: no finite-sample fitting, no compression objective, no under-dimensioned latent space. The positive extreme of what the framework predicts when admissibility is fully specified.

Addition. Explains precisely why the solver works and what its limitation is: the geometric embedding is hand-crafted, so the system cannot discover new invariants or handle transformations outside the declared geometric language.

What we do not claim. We do not claim transversal ARC solving is a model for practical deep learning. Complete domain knowledge of the relevant geometric structure is unavailable in most settings.

Reframing 8. Emergent world models as spontaneous quotient discovery. [Strong compatibility]

Original explanation. Li et al. [2023] trained a GPT variant to predict legal Othello moves from game transcripts, finding that nonlinear probes can recover the board state from internal representations with high accuracy. Gurnee and Tegmark [2024] found analogous spatial and temporal world representations in large language models.

Alignment. The Othello board state is the lawful quotient of the move sequence: two game transcripts reaching the same board position are nuisance-equivalent. A network predicting legal moves must represent which moves are legal, which requires knowing the board state. The board state IS the quotient Q that the prediction task implicitly declares as the relevant structure. This is the positive case the NTK account cannot

accommodate: rich-regime training discovers quotient structure without explicit supervision of Q .

Addition. Explains why positive quotient discovery is possible here—the task-quotient alignment is strong—and provides the distinguishing prediction of section 3.5 (Prediction 2).

What we do not claim. We do not claim that all prediction-trained networks will develop faithful world models. The conditions favouring this outcome are not always satisfied.

Reframing 9. Score-based diffusion models as systematic graded quotient discovery. [Strong structural parallel; topology-faithful convergence not established]

Illustrates: section 3.1 (SSL anti-collapse — architectural escape from graded compression); section 3.2 (rate-distortion and NSW threshold bridge); section 4.2 (Reframings 6–8, positive recovery instances)

Original explanation. Score-based diffusion models [Song et al., 2021, Ho et al., 2020] learn a score function $\nabla_x \log p_t(x)$ at each noise level $\sigma(t)$ of a forward process that adds Gaussian noise to training data. The reverse SDE

$$dx = [-f(x, t) + g^2(t)\nabla_x \log p_t(x)]dt + g(t) d\tilde{W}_t$$

generates samples by reversing the noise process. The training objective is score matching at every noise level simultaneously, and the learned model must faithfully represent data structure at multiple scales to achieve low loss.

Alignment. Score-based diffusion models exhibit three structural properties that align directly with the quotient-faithfulness framework.

First, the noise schedule $\sigma(t)$ implements a systematic traversal of the graded quotient filtration $\{Q_{c,\varepsilon}\}$ established in Paper 2 [Farkas, 2026b] (Theorem 3.5). At high noise, fine-grained semantic distinctions are destroyed and only coarse quotient structure persists; at low noise, fine-grained structure is preserved. The reverse process must therefore encode the full filtration, not merely the coarsest quotient. This is a stronger requirement than any single-scale learning objective.

Second, the forward process constitutes an implicit admissibility declaration. By determining which distinctions are collapsed at each noise scale, the physics of the forward process specifies the admissibility source for each level of the filtration. This source is not architectural (as in equivariant networks) or task-derived (as in contrastive methods) but thermodynamic: it follows from the isotropic Gaussian noise model applied consistently across scales.

Third, and most structurally important, the graded compression of Theorem 4.4 is avoided by design. The score-matching objective at noise level $\sigma(t)$ requires the model to reconstruct the full data distribution at that scale, not to predict a coarser summary. The task map is the identity on $Q_{c,\varepsilon(\sigma(t))}$: there is no non-injective t to create an E_{label} minimum that collapses fine-grained quotient distinctions. This distinguishes diffusion models structurally from classification and contrastive objectives, which are vulnerable to the graded compression whenever the task map collapses semantic distinctions that are not discriminative.

The NSW threshold of Corollary 5.7.2 (Paper 2, [Farkas, 2026b]) provides a formal link

between the noise schedule and topology-faithfulness. The condition

$$\sigma(t) \leq \frac{\sqrt{3/5} \tau}{s_1 \sqrt{2 \log(2d/\delta)}}$$

— where τ is the reach of $\phi(Q)$ and s_1 is the largest singular value of the encoder Jacobian — identifies the noise level below which the reverse process operates in a topology-preserving regime for $Q_{c,\varepsilon(\sigma(t))}$. Above this threshold, topology is destroyed and only coarser structure is recoverable; below it, the full topological structure of the quotient at that scale is in principle accessible.

Addition. Within the framework map of section 3, score-based diffusion models occupy a distinctive position. Against the compression tradition (IB, SSL anti-collapse methods), they demonstrate that the graded compression can be avoided structurally by choosing objectives that preserve rather than compress. Against the SSL anti-collapse tradition specifically, they distinguish dimensional collapse (which anti-collapse methods prevent) from semantic collapse at every scale (which diffusion models address through the multi-scale objective). Against causal representation learning, they provide a mechanism for discovering quotient structure without an explicit causal admissibility declaration — the admissibility is supplied by the noise schedule.

The connection to the Platonic Representation Hypothesis (Reframing 9, section 4.3) is that scale and diversity of training, which Huh et al. identify as the drivers of representation convergence, map onto the diffusion model’s coverage of the full filtration: a model trained on a sufficiently large and diverse dataset must learn the score function at all noise levels, which requires encoding $Q_{c,\varepsilon}$ for the full range of ε . The diffusion model is therefore the strongest known architectural instantiation of the Platonic convergence hypothesis.

What we do not claim. This is a structural parallel, not a derived consequence. Three things are not established. First, the convergence of the learned score function $\hat{s}(x, t)$ to topology-faithful encoding of $Q_{c,\varepsilon(\sigma(t))}$ is not proved; the NSW threshold identifies the condition but does not guarantee convergence. Second, the admissibility declaration implicit in the noise schedule may not match the true nuisance structure of the data-generating process: isotropic Gaussian noise treats all directions equally, which is not always the right admissibility source. Third, the argument concerns the structural properties of the objective; whether any particular trained model achieves the predicted quotient structure depends on training dynamics, architecture capacity, and dataset coverage that are not characterised here. The open problem for Paper 2 (Open Problem 5 of Paper 2, [Farkas, 2026b]) states the convergence question precisely.

Theoretical note: Chain-of-thought reasoning as graded quotient traversal. *The following applies the graded quotient filtration to a class of training procedures. It is a theoretical application rather than a reframing of named empirical evidence and is not counted among the fourteen numbered reframings. [Strong compatibility; Zone classification conjectural]*

Original explanation. Chain-of-thought prompting [Wei et al., 2022] improves reasoning on complex tasks by eliciting intermediate steps before a final answer. Process reward models [Lightman et al., 2024] extend this by training models to evaluate the semantic correctness of each intermediate step. The prevailing explanation is that intermediate steps reduce effective problem complexity and allow models to leverage learned reasoning patterns.

Alignment. In the framework’s language, a CoT trace $x \rightarrow z_1 \rightarrow \dots \rightarrow z_k \rightarrow y$ implements the graded quotient filtration of Paper 2 [Farkas, 2026b]: each step z_i is a partial encoding of the problem state at that reasoning level, defining a natural nested sequence of sub-quotients $Q^{(0)} \supset Q^{(1)} \supset \dots \supset \mathcal{Y}$. This is the Kolmogorov complexity framework applied at scale: each step encodes only the distinctions needed at that level, distributing the embedding burden of the full quotient across the chain.

Three consequences of the framework are immediately interpretable. First, *self-distillation failure* is the Zone 1 graded compression operating at the level of reasoning traces: when the teacher model has compressed to E_{label} , its CoT traces carry no admissibility signal beyond the direct (x, y) supervision, and the student inherits the compression at every step. Second, *process reward models* are the closest existing instantiation of step-level admissibility declarations: a PRM that correctly evaluates intermediate semantic faithfulness is providing Route A structure at each level of the filtration. Third, *latent CoT* (thinking tokens) is the pure case: the latent state at each step must embed $Q^{(i)}$, subject to the dimension floor of Paper 3 [Farkas, 2026c].

Addition. The framework reframes the empirical question “does CoT help?” into a more precise structural question: at which zone of the three-zone compression structure is the reasoning chain operating? A trace that is Zone 1 (semantically empty steps) may improve benchmark performance without traversing the quotient structure. A trace that is Zone 3 (faithful step-level admissibility) would constitute declared admissibility applied to the reasoning process itself. Process reward models with correct semantic evaluation are the path from Zone 1 to Zone 3.

What we do not claim. Whether any training procedure reliably produces Zone 3 traces is open (Paper 2, Open Problem 8). The zone classification of any specific CoT training procedure — including RLVR/GRPO-based methods — requires empirical verification via the quotient-faithfulness diagnostic rather than benchmark performance alone.

4.3 Group III: Scale and convergence

Reframing 10. The Platonic Representation Hypothesis as scale-driven trap weakening. [Strong compatibility; arrival unconfirmed]

Original explanation. Huh et al. [2024] documented that as vision and language models grow larger and are trained on more diverse data and tasks, their internal similarity kernels converge across architectures, training objectives, and modalities. They hypothesise convergence toward a shared statistical model of underlying reality.

Alignment. The three selective pressures they identify map precisely onto the trap taxonomy: task diversity weakens the graded compression (multiple tasks have different E_{label} minima; the only representation achieving low loss across all tasks simultaneously must preserve more of Q ’s structure); data diversity satisfies Proposition 5.6C’s coverage conditions; and scale resolves the embedding floor. The Platonic convergence is the empirical signature of all three failure modes being weakened simultaneously.

Addition. The first mechanistic explanation of why the Platonic convergence occurs. The paper’s authors explicitly acknowledge a mechanistic account in terms of admissibility structure is what the hypothesis leaves implicit: it identifies convergence empirically but does not characterise which equivalence relation the converged representations respect or why any particular quotient is selected.

What we do not claim. Kernel convergence (what Huh et al. measure) is necessary but not sufficient for topology-faithful encoding. Whether large models have fully arrived at

faithful encodings of Q is unconfirmed.

Addition from Dingle–Hutter. Dingle and Hutter [2026] provide the a priori theoretical expectation: when different systems train on simple objectives over simple solution spaces, their optima draw from the same exponentially small set of KC-simple functions, making convergence exponentially more likely than a random-objective null model predicts [Valle-Pérez et al., 2018, Mingard et al., 2025]. The Platonic convergence is therefore not merely empirical regularity—it is the predicted consequence of simplicity bias acting on simple objectives. A critical non-equivalence: convergence toward KC-simple solutions does not certify semantic legitimacy. $E_{\text{label}} = t \circ q$ is KC-simple, and wrongly-collapsed representations can also be KC-simple. Simplicity bias explains convergence; it does not explain what the converged object is in terms of semantic correctness. The admissibility declaration breaks the tie.

Reframing 11. Superposition as embedding-floor competition among quotients. [Compatible observation; dual of the embedding floor]

Original explanation. Elhage et al. [2022] proposed the superposition hypothesis: networks represent more features than they have dimensions by encoding them as nearly-orthogonal directions in activation space. When features are sparse, the interference is small enough that the network can store many more features than it has dimensions.

Alignment. Superposition is the embedding floor operating in an inverse regime. Paper 3’s usual question: is $\dim(Z) \geq \text{emb}(Q)$ for a single known Q ? Superposition’s question: for a given $\dim(Z)$, how many quotients $Q_1, Q_2, \dots$ can the network simultaneously represent? When $\sum_i \text{emb}(Q_i) > \dim(Z)$, faithful encoding of all quotients simultaneously is geometrically impossible. The phase transition between monosemantic and polysemantic regimes is the boundary where the dimensional budget can no longer accommodate all quotients faithfully.

Addition. Provides the structural reason for the monosemantic/polysemantic tradeoff: it is the embedding floor applied to a competition among quotients.

What we do not claim. The extent and character of superposition in frontier models remains an active research question.

Reframing 12. Mode connectivity as parameter-space heterogeneity evidence. [Compatible observation; function-space distinction proved by Proposition 5.6P]

Original explanation. Draxler et al. [2018] and Garipov et al. [2018] showed that many independently trained solutions can be connected by low-loss paths after weight permutation alignment. Liao et al. [2024] found RSB-like fracturing in wide networks, with solutions organising into hierarchical non-isomorphic clusters.

Alignment. Proposition 5.6P establishes that traversal-type solutions and quotient-faithful solutions cannot be continuously deformed into one another in function space without crossing $\Sigma(Q, Z)$. Mode connectivity failure is consistent with pairs of solutions being on opposite sides of this discriminant.

What we do not claim. The translation from function-space disconnection to parameter-space connectivity failure depends on how the parameterisation map folds parameter space, and this step does not close.

4.4 Group IV: The traversal-as-feature inversion

Reframing 13. Speculative and speculative-speculative decoding as quotient traversal with faithful verification. [Structural parallel; mechanism differs]

Original explanation. Speculative decoding [Leviathan et al., 2023, Chen et al., 2023] accelerates inference by using a fast draft model to generate candidate token sequences, verified by the target model. The acceptance rate equals $1 - \frac{1}{2}\|p_{\text{target}} - p_{\text{draft}}\|_1$. Speculative speculative decoding [Kumar et al., 2026] removes the remaining sequential bottleneck by pre-speculating multiple likely verification outcomes in parallel.

Alignment. The draft model operates in a coarser semantic space than the target model, traversing the target’s quotient at lower resolution. The target model’s verification step is the faithful quotient check. The structural inversion: in Papers 1 and 2, traversal is the failure mode. Here, traversal is the mechanism that makes the system fast—because a faithful verifier corrects its errors. The finite-sample traversal is a trap precisely when there is no verifier.

Addition. Illustrates that the traversal/fidful distinction is contextually relative to whether a verifier exists.

What we do not claim. The draft model is not designed as a quotient encoder of the target; its lower dimensionality is a consequence of being smaller, not an explicit architectural choice about equivalence relations.

Reframing 14. Lottery tickets as evidence for distributed fragility. [Compatible observation; Proposition 5.6P resolves one of two required conjectures]

Original explanation. Frankle and Carlin [2019] showed that dense trained networks contain sparse subnetworks that, when retrained from their original initialisation values, match full-network performance. These subnetworks fail to train when randomly reinitialised to new values despite having the same topology.

Alignment. Proposition 5.6P establishes that any training trajectory from a traversal-type solution to a quotient-fidful embedding must cross $\Sigma(Q, Z)$. Random reinitialisation places the winning-ticket subnetwork on the traversal side of $\Sigma(Q, Z)$; the sparse topology’s reduced gradient signal may not be sufficient to drive the required crossing from a new starting point. The distributed encoding of quotient structure over the full circuit—not localisable in the sparse topology alone—is the point.

What we do not claim. The first conjecture—that the winning subnetwork specifically encodes quotient structure rather than arbitrary task-predictive features—remains interpretive.

5 Open Problems as Reframing Problems

5.1 The certification gap: when can quotient faithfulness be verified?

The framework can characterise what faithful encoding means and the obstacles to achieving it, but has no general method for verifying whether a given trained representation has achieved it. Theorem 3.7 (Paper 2) converts certification into a learning problem: if the semantic ultrapseudometric d_c can be estimated and verified at scale ε , the hard subquotient $Q_{c,\varepsilon}$ is certified. Two questions remain open: whether a learned d_c converges to the true admissibility structure, and whether the convergent d_c is identifiable up to the symmetries of the problem.

The mirage reasoning reframing illustrates the stakes: frontier models can achieve top benchmark performance while encoding no visual quotient structure, and standard evaluation protocols cannot detect this failure. A certification result—stating conditions under which a verification procedure certifies faithfulness with probability $\geq 1 - \delta$ —does not currently exist for any nontrivial quotient Q .

5.2 Metric enforcement: when topology is not enough

The framework’s primary faithfulness condition is topological: $\tilde{E} : Q \rightarrow Z$ must be a homeomorphism. This places no constraint on the metric structure of the encoding. For downstream uses—linear interpolation between representations, distance-based retrieval, nearest-neighbour classification—metric faithfulness is required.

Corollary 5.7.3 (Paper 2) provides the bridge: the NSW threshold $\sigma \leq \sqrt{3/5} \cdot \tau / (s_1 \sqrt{2 \log(2d/\delta)})$ connects topologically tolerable noise to the reach τ of Q and the Jacobian spectrum. The Sahraee-Ardakan et al. (2026) diffusion result provides a concrete falsifiable prediction: velocity-parameterised diffusion models should produce metrically less distorted representations near the data manifold than noise-prediction parameterisations, independently of topological faithfulness.

The metric-floor side of this gap is now developed in Farkas [2026c] §3.4: a bi-Lipschitz floor at distortion D is controlled by the doubling dimension up to constants (Naor–Neiman 2012); the C^∞ -smooth floor is Whitney’s classical $2n$; the C^1 -isometric floor is $n + 1$ (Nash–Kuiper); the C^∞ -isometric floor is $O(n^2)$ (Nash–Günther). These are necessary conditions and existence statements for restricted classes — doubling metric spaces, smooth compact manifolds. What remains open on the upper-bound side is the operative question: metric-faithful upper bounds in generality and, more urgently, in the learning setting. Classical isometric embedding theorems are existence results; they do not guarantee that a network of the permitted width, trained by gradient descent, will find a metric-faithful encoding. The operative-contract selection problem (Farkas [2026c], §12.5) and the discovery-versus-existence gap (§5.3) together cover this second half.

5.3 Discovery upper bounds: from existence to learnability

Paper 3 establishes that faithful encoding of Q exists in a space of dimension $\geq \text{emb}(Q)$. It says nothing about whether gradient descent finds such an encoding. A discovery upper bound would take the form: if the data distribution satisfies condition $\mathcal{D}$, the architecture satisfies $\mathcal{A}$, and the objective satisfies $\mathcal{O}$, then with probability $\geq 1 - \delta$, a network of width $\text{emb}(Q)$ converges to a topology-faithful encoding of Q within T steps. No such result currently exists for any non-trivial Q .

The transversal ARC solver (Reframing 7) is the positive extreme: when admissibility is completely specified, all three failure modes are avoided and faithful encoding is achieved by construction. The mirage reasoning finding (Reframing 2) is the negative extreme. The discovery upper bounds problem asks what conditions produce outcomes closer to the former than the latter.

Speculative decoding (Reframing 12) reveals a fourth logically independent problem: efficient traversal of a learned faithful representation at inference time. Once the faithful encoding exists, traversal of it by a coarser quotient model becomes the engineering problem—the traversal-as-feature inversion.

6 What Would Falsify These Claims?

A perspective paper that cannot be falsified is advocacy, not science. This section states the experimental conditions under which each major claim of this paper would be confirmed or refuted.

6.1 Framework-level falsifiers

The mathematical core of the framework—Theorem 4.6, Theorem 4.4, and Corollary 3.6—consists of proved theorems and cannot be empirically falsified. What can be falsified are the practical claims the framework makes about when these results bind in realistic training settings.

F1. The graded compression is dynamically non-binding at sufficient width.

Theorem 4.4 establishes that the global minimum of the compression objective when t is non-injective on Q is $E_{\text{label}} = t \circ q$. This is a claim about the loss landscape, not the training trajectory. The practical claim is that training reaches this minimum.

Falsifier. A model trained on task loss alone—without topology losses, metric constraints, equivariant architecture, noise regularisation, or any other form of structural regularisation—that achieves verified topology-faithful encoding of Q in a setting where t is demonstrably non-injective on Q and $\dim(Z) \geq \text{emb}(Q)$. Faithfulness must be verified mechanistically, not inferred from task accuracy.

What this would mean. Theorem 4.4 remains correct. What would be falsified is the practical claim that gradient descent reaches E_{label} . This would require adding overparameterisation as a fifth recovery mechanism for the graded compression.

Current status. The mirage reasoning finding (Reframing 2) shows the graded compression binding at frontier scale in the multimodal setting; the mLaSDI finding (Reframing 1) shows it binding in scientific ML. Indirect pressure from the implicit bias and lazy-to-rich grokking literature exists, but none satisfies the required mechanistic verification criterion. Existing no-regularisation generalisation results typically do not verify topology-faithful recovery of a declared non-injective quotient.

F2. The batch-instability asymmetry is not the primary mechanism driving traversal escape.

Propositions 5.6A, 5.6C; Lemma 5.6D; and Proposition 5.6P together constitute a mechanistic account of traversal escape with specific, batch-size-dependent predictions.

Falsifier (three separable sub-predictions).

- *F2a (from Proposition 5.6A):* Traversal-trap escape timing should scale with batch size as $(N - b)/b$. This is a quantitative prediction with no free parameters once $S(\theta)$ is measured.
- *F2b (from Proposition 5.6C):* $S(\theta_{\text{trav}}) > S(\theta_{\text{quot}})$ at known traversal and quotient-aligned parameter configurations, with the decomposition $W(\theta) + B(\theta)$ matching the named coverage conditions. Directly measurable in controlled settings.
- *F2c (from Conjecture 5.6E):* Escape events correlate with mini-batch episodes where the batch-conditioned projected curvature $u^\top \nabla^2 R_B(\theta_{\text{trav}})u \leq 0$. Testable from training traces.

What refutation would mean. If escape timing does not scale with batch size as Proposition 5.6A predicts, or if $S(\theta_{\text{trav}}) \leq S(\theta_{\text{quot}})$ in controlled measurements, the batch-

instability account is wrong as the primary mechanism.

Current literature. Existing grokking work provides genuine competing mechanisms centred on weight decay, norm dynamics, and numerical stability [Nanda et al., 2023, Barak et al., 2025]. No published study tests the full three-part batch-instability signature proposed here. The Softmax Collapse account and our $S(\theta)$ account are mechanistically complementary—SC prevention removes a numerical barrier; $S(\theta)$ asymmetry provides the structural driving force—and may both operate simultaneously.

F3. The final latent space dimension floor is not architecturally binding in practice. Corollary 3.6 establishes that no topology-faithful encoding of Q exists when $\dim(Z) < \text{emb}(Q)$. The practical claim is that encoders whose final latent space falls below $\text{emb}(Q)$ systematically fail to achieve topology-faithful encoding of Q .

Falsifier. A topology-faithful encoding of a known non-trivial quotient Q , verified mechanistically against the final encoder output Z (not any intermediate layer), achieved with $\dim(Z) < \text{emb}(Q)$. The quotient must be correctly declared—not a lower-dimensional approximation that happens to embed in the smaller space.

What this would mean. The theorem remains correct. A genuine faithful encoding below the floor is mathematically impossible; what would have occurred is either a faithfulness test measuring something weaker than the formal topological condition, or a misidentification of Q . Both outcomes are scientifically useful. The former would establish that the formal floor is too strong for the practically relevant target, motivating a shift to the graded faithfulness framework at coarser semantic scales; the latter would require revising the admissibility declaration.

Current literature. Published literature on topology-aware autoencoders and homeomorphic encoders is broadly consistent with the architectural bindingness of the dimension floor—nontrivial topology imposes real constraints, and atlas/multi-chart methods succeed by changing the represented object (decomposing into local patches) rather than circumventing the global encoding constraint.

6.2 The three distinguishing predictions

The three predictions from section 3.5 distinguish the framework from specific competing accounts:

Prediction 1 (grokking batch-size scaling) distinguishes the $S(\theta)$ account from the norm-penalty account. Confirming outcome: transition timing scales with $(N - b)/b$; $S(\theta_{\text{trav}})$ measurably exceeds $S(\theta_{\text{quot}})$. Refuting outcome: timing is independent of batch size but dependent on weight decay magnitude.

Prediction 2 (kernel-regime quotient faithfulness) distinguishes our framework from the NTK account. Confirming outcome: rich-regime networks show mechanistic evidence of quotient encoding at matched task accuracy; lazy-regime networks do not. Refuting outcome: kernel-regime networks show equivalent mechanistic evidence—the NTK’s fixed features happen to be aligned with Q .

Prediction 3 (task accuracy independence) distinguishes our framework from accounts that treat task accuracy as a proxy for representational quality. Confirming outcome: task-accurate models show variable quotient faithfulness uncorrelated with benchmark accuracy; the mirage reasoning finding is the existence proof. Refuting outcome: task accuracy and quotient faithfulness are strongly correlated at sufficient scale—which would strengthen the Platonic hypothesis significantly.

6.3 What confirmation would look like

The single experiment that would most strongly confirm the framework’s central claims: a controlled study that simultaneously tests all three traps—compression, traversal, and dimension—across the same set of models and tasks, showing that the three failure modes are empirically separable, each responds to the predicted remedy, and together they account for the variance in representational quality across models. No such experiment currently exists. Designing and executing it would be the single most valuable empirical contribution to this research programme.

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- Peter Farkas. Admissibility sources and the graded compression: Three routes to quotient recovery. Preprint, 2026b. Paper II in the series *Quotient Faithfulness in Representation Learning*.
- Peter Farkas. Dimension floors for quotient-faithful encoding: Architectural bounds in representation learning. Preprint, 2026c. Paper III in the series *Quotient Faithfulness in Representation Learning*.
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A Definitions and Key Concepts

Paper 5 introduces no new formal definitions. It uses the following key conceptual terms, each defined formally in the cited paper.

Admissibility declaration. An explicit specification of $\sim$ from any of the four sources of Paper 2 [Farkas, 2026b].

Three traps. Traversal (Thm. 4.6), Compression (Thm. 4.4), Dimension (Cor. 3.6) of Paper I [Farkas, 2026a].

Three recovery routes. Route A (declared admissibility), Route A-extension (incentivised discovery), Route B (stochastic escape) of Paper 2 [Farkas, 2026b].

Three-zone compression structure. Zone 1 (no escape), Zone 2 (partial via regulariser), Zone 3 (full via declared admissibility); Paper 2 [Farkas, 2026b], §6.

Evidential labels. Five-tier scheme for reframings: Direct structural evidence, Strong compatibility, Compatible observation, Structural parallel, Mechanism differs.

B Contribution Attribution and AI Disclosure

B.1 Context

Paper 5 was developed in extended sessions with Claude Sonnet 4.6 (Anthropic, Apr. 2026) under the intellectual direction of the first author, with editorial and calibration input from ChatGPT 5.4 (OpenAI), Gemini 3 (Google), and MetaAI (Meta) across the series. The attribution system follows Papers I–IIb: ++ for unsolicited AI contributions by any of the tools used; + for directed contributions; 0 for non-AI work. The AI column records the maximum weight across AI tools for a given row.

The first author takes full scientific responsibility for all claims.

B.2 Contribution table (reframing-relevant entries)

Contribution	Character	Lit.	Autho	AI
Series framing and paper structure	novel	0	++	0
Framework map — six traditions	novel synthesis	+	++	++

Contribution	Character	Lit.	Autho	AI
Chain 1 (invariance) and Chain 2 (geometry) intellectual lineage	synthesis	0	++	0
GDL as seed, gap identification	reframing	+	++	0
IB as parent framework with $\sim$ as missing structural component	reframing	+	++	++
SSL anti-collapse: dimensional vs. semantic collapse distinction	novel	+	++	++
Causal RL: trap taxonomy as the missing obstacle characterisation	synthesis	+	++	+
Disentanglement: quotient faithfulness as achievable target	reframing	+	++	+
Flat minima $\rightarrow S(\theta)$ formalisation as Chain 2	synthesis	0	++	+
SLT / RLCT as complementary instrument	synthesis	+	++	++
MoE as local quotient covering (atlas interpretation)	novel	0	++	++
Implicit bias family: opposed on graded compression	reframing	+	++	+
PAC-Bayes: overparameterisation revision	synthesis	+	++	+
Mechanistic interpretability: complementarity via Chughtai et al.	synthesis	+	++	+
Fourteen reframings (all)	novel	+	++	++
	reframing			
Platonic hypothesis: mechanistic account + Dingle–H. convergence addition	novel	0	++	++
SD/SSD: traversal-as-feature inversion	novel	+	++	++
	reframing			
F1, F2, F3 falsifier formulation	novel	+	++	+
Hostile literature map	analysis	+	++	++
§6.1 current literature acknowledgements	analysis	+	++	+
Abstract, §1, §2 prose	writing	+	++	++
<i>Additions: Apr. 2026</i>				
Theoretical note: CoT as graded quotient traversal	novel	+	++	+
	reframing			
Dingle–H. as seventh adjacent tradition (§3.2)	novel syn-	++	+	+
	thesis			
Grokking soliton framing in Reframing 3	novel fram-	+	++	+
	ing			
Tone audit: six §3.1 tradition analyses revised	editorial	0	++	+
“What this paper does not claim” paragraph	structural	0	++	+

Contribution	Character	Lit.	Author	AI
Notation page, definitions appendix, structural	structural	0	+	++

C Attribution Methodology

The attribution methodology (the ++/+/0 weighting scheme and its worked example) is documented in full in Paper I [Farkas, 2026a], Appendix D, and Paper 2 [Farkas, 2026b], Appendix D. The same scheme is applied in this paper; readers seeking the full prompt and methodology should consult those appendices.